CM5 socket, M.2 connector

File: cm5_socket.kicad_sch

System MCU, Watchdog, EEPROM

File: system_mcu.kicad_sch

AC power, DC power, AC sync

File: ac_dc_power.kicad_sch

Isolated digital IO, Fan controller, Encoder

File: digital_io_fan_encoder.kicad_sch

7-port USB3 Hub

File: usb3_hub.kicad_sch

Rear USB, USB-C

File: usb_conn.kicad_sch

Ethernet, Video outputs, Ref clk

File: video_outputs.kicad_sch

DIB 2.0

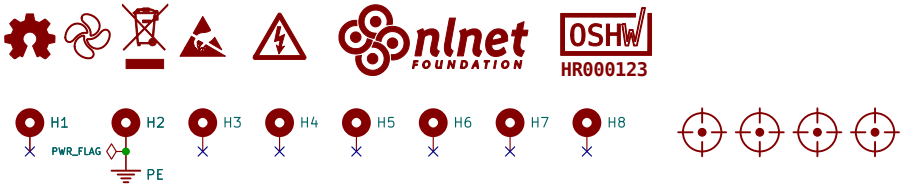
File: dib.kicad_sch

Trigger switch matrix, Power modules coupling

File: power_coupling.kicad_sch

USB Port	USB type	STM32H7 CM5	Raspberry PI CM5
Rear top	USB A	FS only	USB #0, HS/SS
Rear bottom	USB A	FS/HS	USB Hub P1 Flex
Front	USB C	USB Hub P2, HS/SS, Dual role	
DIB #1	DIB 2.0	USB Hub P7, HS/SS	
DIB #2	DIB 2.0	USB Hub P6, HS/SS	
DIB #3	DIB 2.0	USB Hub P5, HS/SS	
DIB #4	DIB 2.0	USB Hub P4, HS/SS	
DIB #5	DIB 2.0	USB Hub P3, HS/SS	

I2C address	Device
0x53	EEPROM M24C32
0x2F	Fan controller EMC2301
0x2D	USB3 Hub
0x3A	PTN5150A USB CC controller
0x70	ADG2188Y 8 x 8 switch array

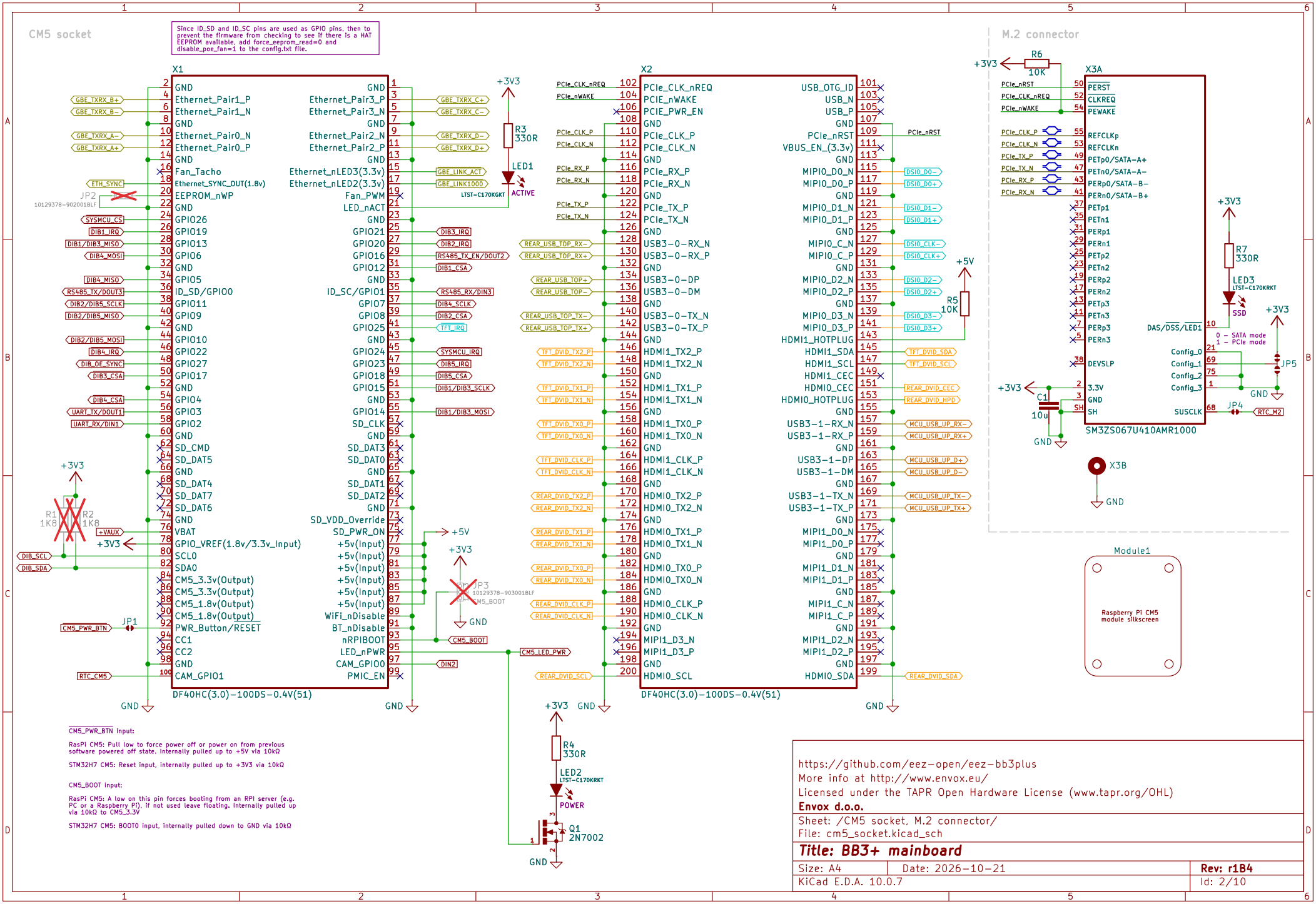


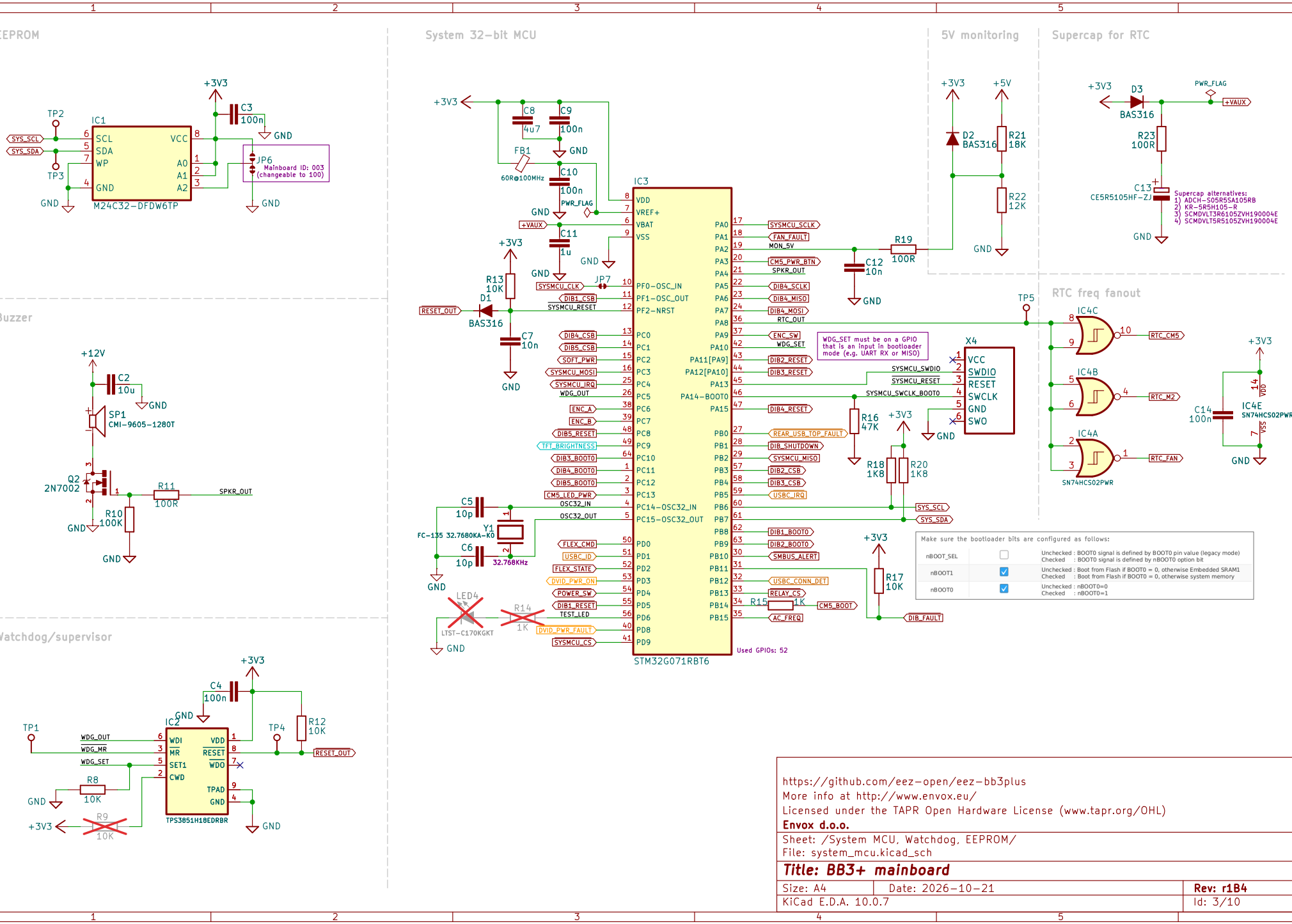
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.

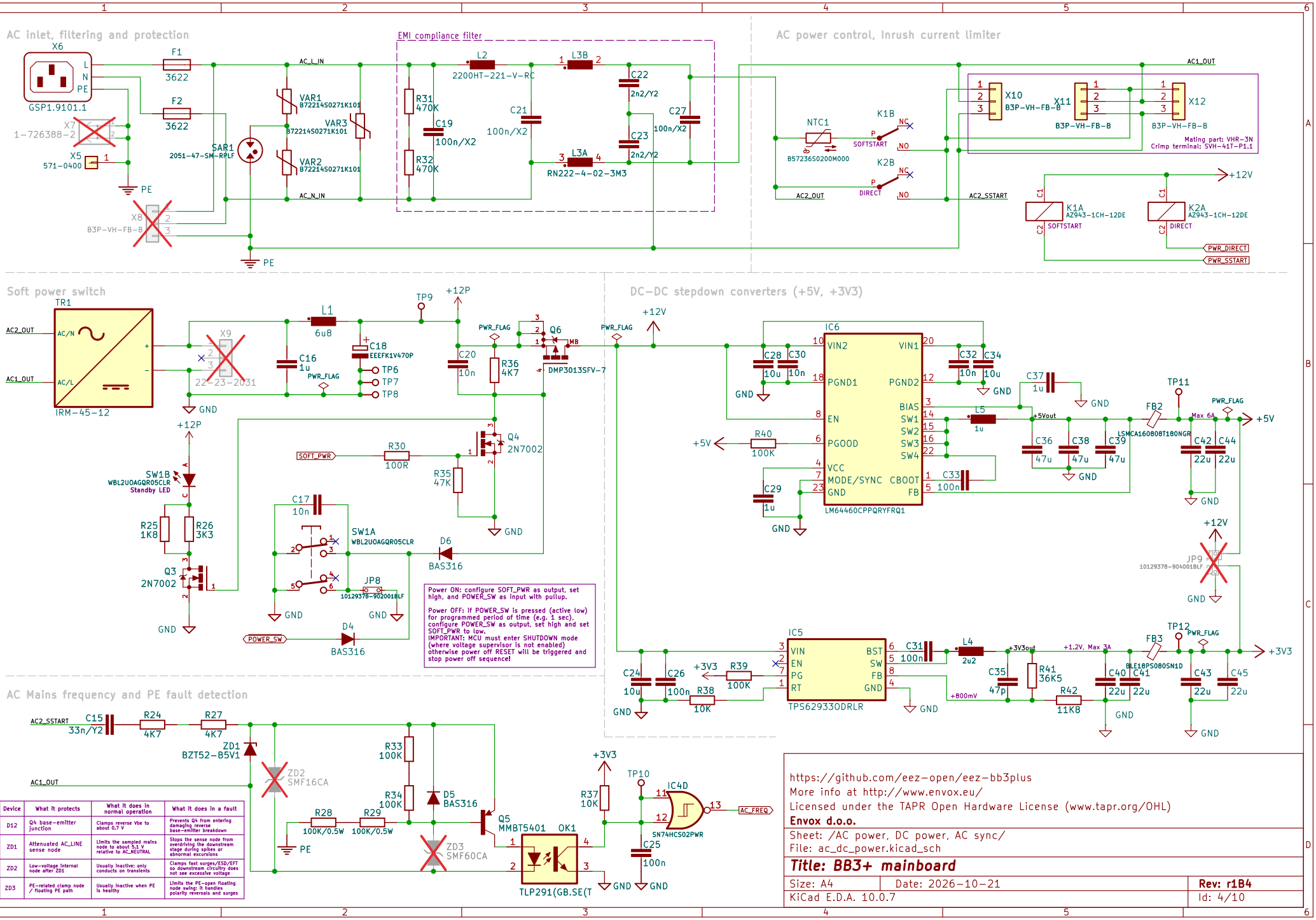
Sheet: /
File: BB3plus mainboard.kicad_sch

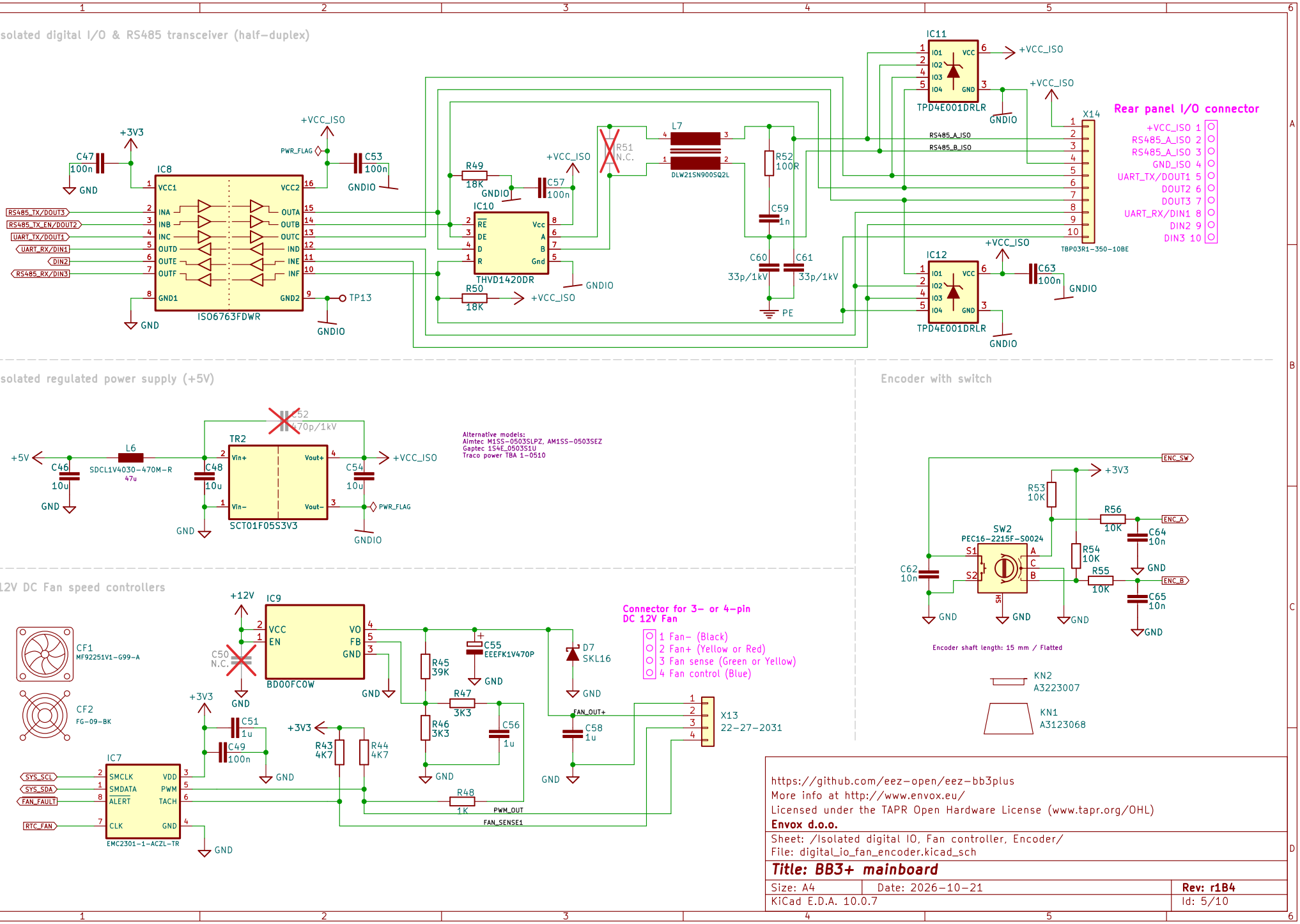
Title: BB3+ mainboard

Size: A4	Date: 2026-10-21	**Rev: r1B4**
KiCad E.D.A. 10.0.7		Id: 1/10


CM5 socket
Since ID_SD and ID_SC pins are used as GPIO pins, then to prevent the firmware from checking to see if there is a HAT EEPROM available, add force_eeprom_read=0 and disable_poe_fan=1 to the config.txt file.
X1
DF40HC(3.0)-100DS-0.4V(51)
X2
DF40HC(3.0)-100DS-0.4V(51)
M.2 connector
X3A
SM3ZS067U410AMR1000
X3B
Module1
Raspberry PI CM5 module silkscreen
R1 1K8
R2 1K8
R3 330R
R4 330R
R5 10K
R6 10K
R7 330R
C1 10u
LED1 ACTIVE
LED2 POWER
LED3 SSD
Q1 2N7002
JP1
JP2
JP3
JP4
JP5
CM5_PWR_BTN input:
RasPI CM5: Pull low to force power off or power on from previous software powered off state. Internally pulled up to +5V via 10kΩ
STM32H7 CM5: Reset input, internally pulled up to +3V3 via 10kΩ
CM5_BOOT input:
RasPi CM5: A low on this pin forces booting from an RPI server (e.g. PC or a Raspberry PI). If not used leave floating. Internally pulled up via 10kΩ to CM5_3.3V
STM32H7 CM5: BOOT0 input, internally pulled down to GND via 10kΩ
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.
Sheet: /CM5 socket, M.2 connector/
File: cm5_socket.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-21
Rev: r1B4
KiCad E.D.A. 10.0.7
Id: 2/10

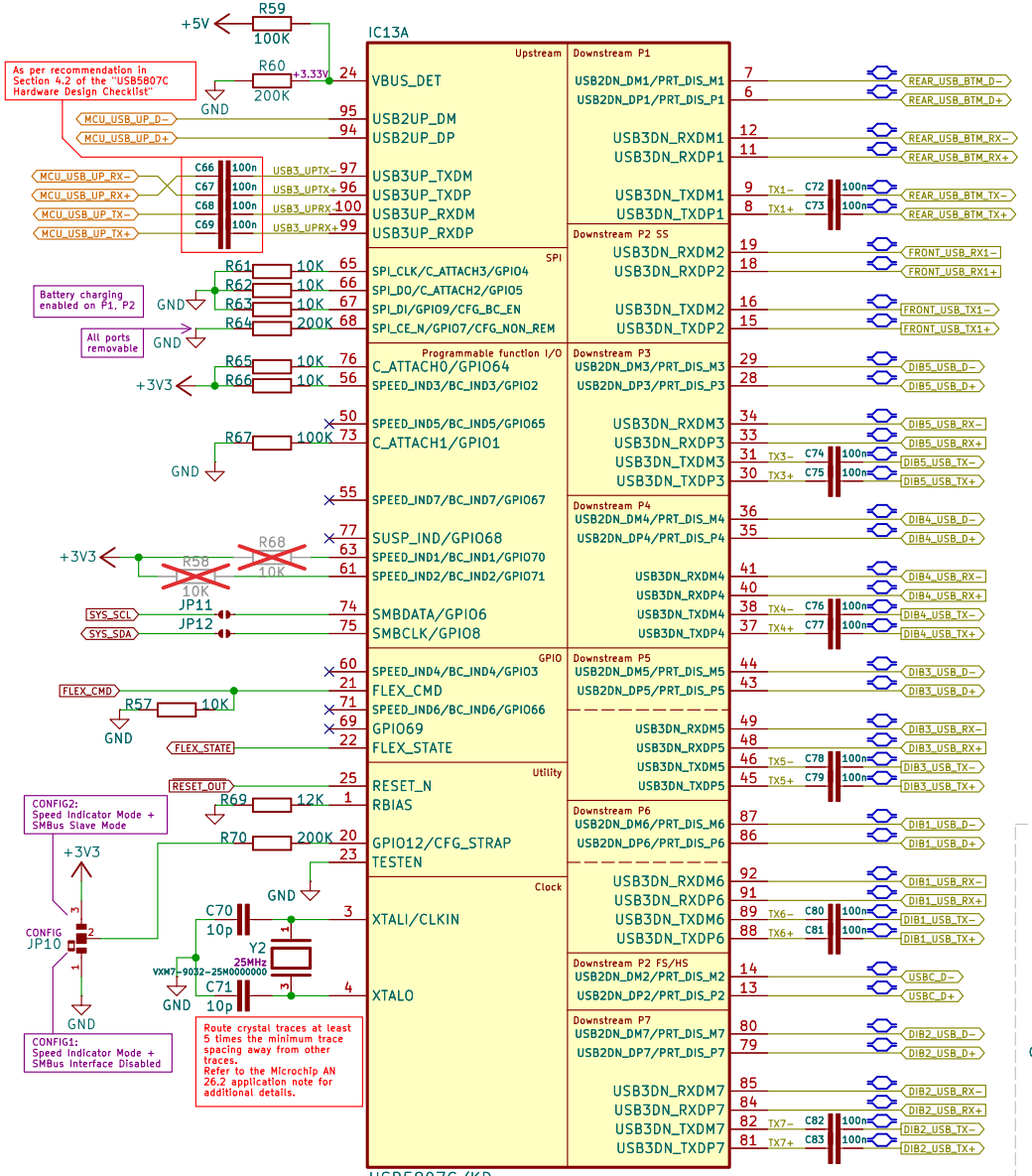

EEPROM
IC1
M24C32-DFDW6TP
TP2
TP3
C3
100n
JP6
Mainboard ID: 003
(changeable to 100)
Buzzer
SP1
CMI-9605-1280T
Q2
2N7002
R10
100K
R11
100R
SPKR_OUT
Watchdog/supervisor
IC2
TPS3851H18EDRBR
C4
100n
R8
10K
R9
10K
R12
10K
TP1
TP4
System 32-bit MCU
IC3
STM32G071RBT6
C8
4u7
C9
100n
C10
100n
FB1
60R@100MHz
C11
1u
C7
10n
D1
BAS316
R13
10K
JP7
C5
10p
C6
10p
Y1
FC-135 32.7680KA-K0
32.768KHz
LED4
LTST-C170KGKT
R14
1K
Used GPIOs: 52
5V monitoring
D2
BAS316
R21
18K
R22
12K
R19
100R
C12
10n
Supercap for RTC
D3
BAS316
R23
100R
C13
CE5R5105HF-ZJ
Supercap alternatives:
1) ADCH-S05R5SA105RB
2) KR-5R5H105-R
3) SCMDVLT3R6105ZVH190004E
4) SCMDVLT5R5105ZVH190004E
RTC freq fanout
IC4C
IC4B
IC4A
SN74HCS02PWR
IC4E
SN74HCS02PWR
C14
100n
X4
SWDIO
RESET
SWCLK
GND
SWO
R16
47K
R18
1K8
R20
1K8
R17
10K
R15
1K
TP5
WDG_SET must be on a GPIO that is an input in bootloader mode (e.g. UART RX or MISO)
Make sure the bootloader bits are configured as follows:
nBOOT_SEL
nBOOT1
nBOOT0
Unchecked : BOOT0 signal is defined by BOOT0 pin value (legacy mode)
Checked : BOOT0 signal is defined by nBOOT0 option bit
Unchecked : Boot from Flash if BOOT0 = 0, otherwise Embedded SRAM1
Checked : Boot from Flash if BOOT0 = 0, otherwise system memory
Unchecked : nBOOT0=0
Checked : nBOOT0=1
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.
Sheet: /System MCU, Watchdog, EEPROM/
File: system_mcu.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-21
Rev: r1B4
KiCad E.D.A. 10.0.7
Id: 3/10


AC inlet, filtering and protection
EMI compliance filter
AC power control, Inrush current limiter
Soft power switch
DC–DC stepdown converters (+5V, +3V3)
AC Mains frequency and PE fault detection
X6
GSP1.9101.1
F1
3622
F2
3622
X7
1–726388–2
X5
571–0400
X8
B3P–VH–FB–B
SAR1
2051–47–SM–RPLF
VAR1
B72214S0271K101
VAR3
B72214S0271K101
VAR2
B72214S0271K101
AC_L_IN
AC_N_IN
PE
R31
470K
R32
470K
C19
100n/X2
L2
2200HT–221–V–RC
C21
100n/X2
L3B
L3A
RN222–4–02–3M3
C22
2n2/Y2
C23
2n2/Y2
C27
100n/X2
NTC1
B57236S0200M000
K1B
K2B
SOFTSTART
DIRECT
AC2_OUT
AC2_SSTART
AC1_OUT
X10
X11
X12
B3P–VH–FB–B
Mating part: VHR–3N
Crimp terminal: SVH–41T–P1.1
K1A
AZ943–1CH–12DE
K2A
AZ943–1CH–12DE
+12V
PWR_DIRECT
PWR_SSTART
TR1
IRM–45–12
AC2_OUT
AC1_OUT
X9
22–23–2031
L1
6u8
C16
1u
C18
EEEFK1V470P
TP6
TP7
TP8
TP9
+12P
C20
10n
R36
4K7
PWR_FLAG
Q6
DMP3013SFV–7
GND
+12P
SW1B
WBL2UOAGQR05CLR
Standby LED
R25
1K8
R26
3K3
Q3
2N7002
SOFT_PWR
R30
100R
Q4
2N7002
R35
47K
C17
10n
SW1A
WBL2UOAGQR05CLR
JP8
10129378–902001BLF
D6
BAS316
D4
BAS316
POWER_SW
Power ON: configure SOFT_PWR as output, set high, and POWER_SW as input with pullup.
Power OFF: if POWER_SW is pressed (active low) for programmed period of time (e.g. 1 sec), configure POWER_SW as output, set high and set SOFT_PWR to low.
IMPORTANT: MCU must enter SHUTDOWN mode (where voltage supervisor is not enabled) otherwise power off RESET will be triggered and stop power off sequence!
IC6
LM64460CPPQRYFRQ1
VIN2
VIN1
PGND1
PGND2
BIAS
EN
SW1
SW2
SW3
SW4
PGOOD
VCC
MODE/SYNC
CBOOT
GND
FB
C28
10u
C30
10n
C32
10n
C34
10u
C37
1u
C29
1u
C33
100n
L5
1u
+5Vout
FB2
LSMCA160808T180NGR
Max 6A
TP11
+5V
R40
100K
C36
47u
C38
47u
C39
47u
C42
22u
C44
22u
JP9
10129378–904001BLF
IC5
TPS629330DRLR
VIN
EN
PG
RT
BST
SW
FB
GND
C24
10u
C26
100n
R39
100K
R38
10K
C31
100n
L4
2u2
+3V3out
+1.2V, Max 3A
FB3
BLE18PS080SN1D
TP12
+3V3
C35
47p
R41
36K5
R42
11K8
+800mV
C40
22u
C41
22u
C43
22u
C45
22u
Device
What it protects
What it does in normal operation
What it does in a fault
D12
Q4 base–emitter junction
Clamps reverse Vbe to about 0.7 V
Prevents Q4 from entering damaging reverse base–emitter breakdown
ZD1
Attenuated AC_LINE sense node
Limits the sampled mains node to about 5.1 V relative to AC_NEUTRAL
Stops the sense node from overdriving the downstream stage during spikes or abnormal excursions
ZD2
Low–voltage internal node after ZD1
Usually inactive; only conducts on transients
Clamps fast surges/ESD/EFT so downstream circuitry does not see excessive voltage
ZD3
PE–related clamp node / floating PE path
Usually inactive when PE is healthy
Limits the PE–open floating node swing; it handles polarity reversals and surges
AC2_SSTART
C15
33n/Y2
R24
4K7
R27
4K7
ZD1
BZT52–B5V1
ZD2
SMF16CA
R33
100K
R34
100K
D5
BAS316
R28
100K/0.5W
R29
100K/0.5W
ZD3
SMF60CA
Q5
MMBT5401
OK1
TLP291(GB.SE(T
R37
10K
TP10
IC4D
SN74HCS02PWR
C25
100n
AC_FREQ
https://github.com/eez–open/eez–bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.
Sheet: /AC power, DC power, AC sync/
File: ac_dc_power.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026–10–21
Rev: r1B4
KiCad E.D.A. 10.0.7
Id: 4/10


Isolated digital I/O & RS485 transceiver (half-duplex)
IC8
ISO6763FDWR
IC10
THVD1420DR
IC11
TPD4E001DRLR
IC12
TPD4E001DRLR
L7
DLW21SN900SQ2L
X14
TBP03R1-350-10BE
Rear panel I/O connector
+VCC_ISO 1
RS485_A_ISO 2
RS485_A_ISO 3
GND_ISO 4
UART_TX/DOUT1 5
DOUT2 6
DOUT3 7
UART_RX/DIN1 8
DIN2 9
DIN3 10
Isolated regulated power supply (+5V)
TR2
SCT01F05S3V3
L6
SDCL1V4030-470M-R
Alternative models:
Aimtec M1SS-0503SLPZ, AM1SS-0503SEZ
Gaptec 1S4E_0503S1U
Traco power TBA 1-0510
12V DC Fan speed controllers
IC9
BD00FC0W
IC7
EMC2301-1-ACZL-TR
CF1
MF92251V1-G99-A
CF2
FG-09-BK
Connector for 3- or 4-pin DC 12V Fan
1 Fan- (Black)
2 Fan+ (Yellow or Red)
3 Fan sense (Green or Yellow)
4 Fan control (Blue)
X13
22-27-2031
Encoder with switch
SW2
PEC16-2215F-S0024
Encoder shaft length: 15 mm / Flatted
KN2
A3223007
KN1
A3123068
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.
Sheet: /Isolated digital IO, Fan controller, Encoder/
File: digital_io_fan_encoder.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-21
Rev: r1B4
KiCad E.D.A. 10.0.7
Id: 5/10

7-port USB hub

All downstream ports that are not used should be disabled (use I2C to configure).

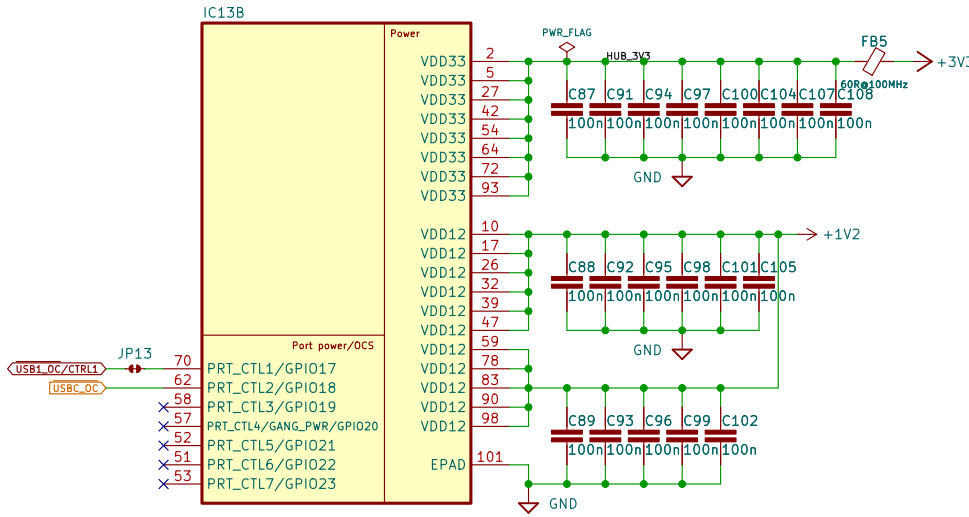


Microchip HUB troubleshoot articles:

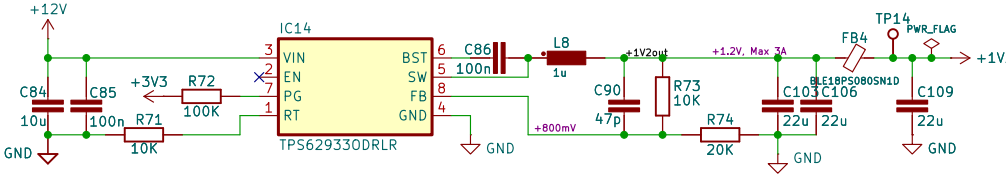
https://support.microchip.com/s/article/USB3-Hub-Upstream-Port-Connect-Enumeration-Failure
https://support.microchip.com/s/article/USB-High-Speed-Chirp-Explained
https://support.microchip.com/s/article/USB3-Connection-Process-and-Attach-Detection

IMPORTANT:

If I2C/SMBus pull-up resistors are detected by the USB5807C at start-up, the hub will wait indefinitely to be configured by the attached I2C/SMBus master. For early prototyping, it may be necessary to physically remove the pull-up resistors until the I2C/SMBus master is fully operational and able to properly configure the hub at start-up.



USB Hub DC-DC stepdown converter (+1V2)

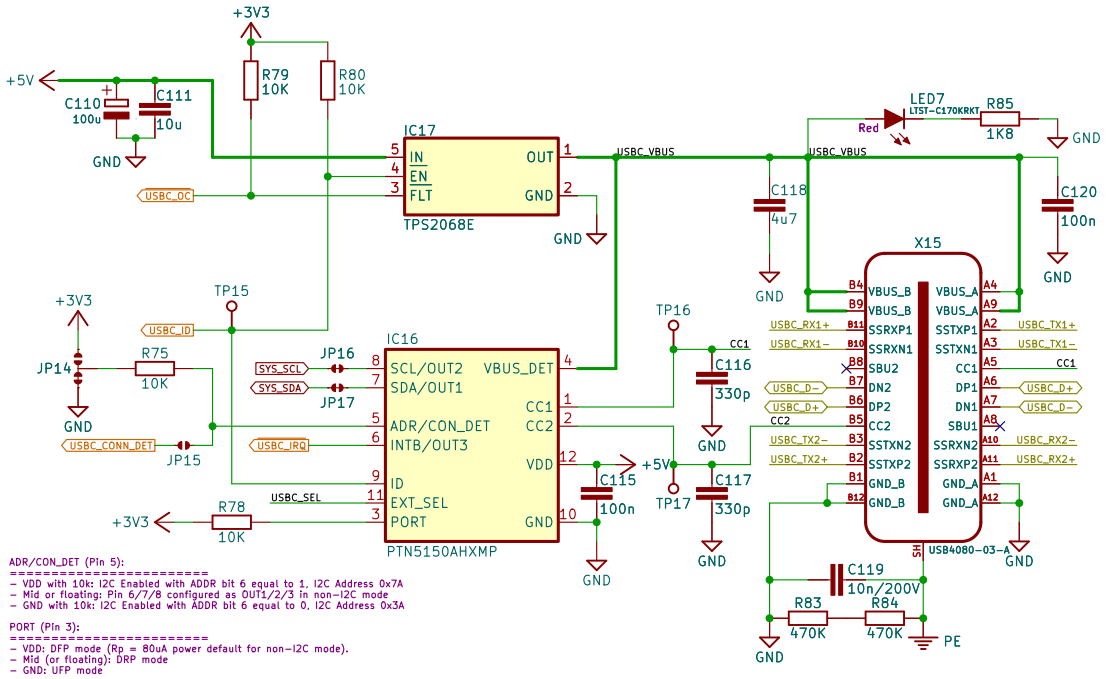


https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

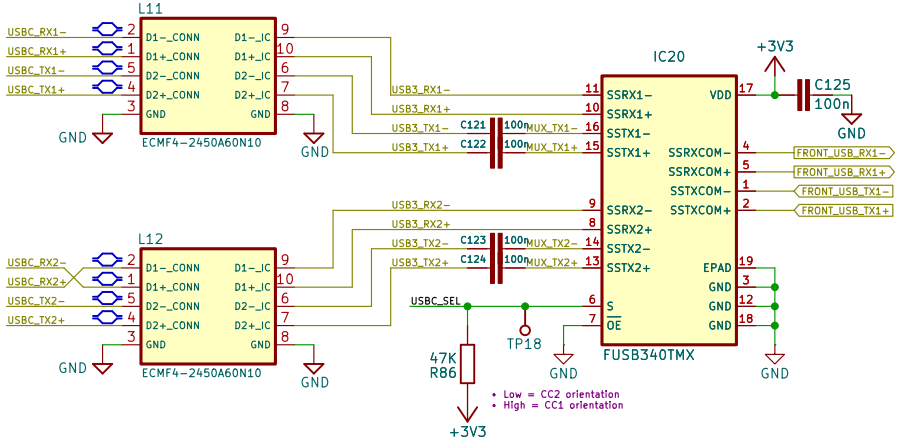
Envox d.o.o.

Sheet: /7-port USB3 Hub/		
File: usb3_hub.kicad_sch		
Title: BB3+ mainboard		
Size: A4	Date: 2026-10-21	**Rev: r1B4**
KiCad E.D.A. 10.0.7		Id: 6/10

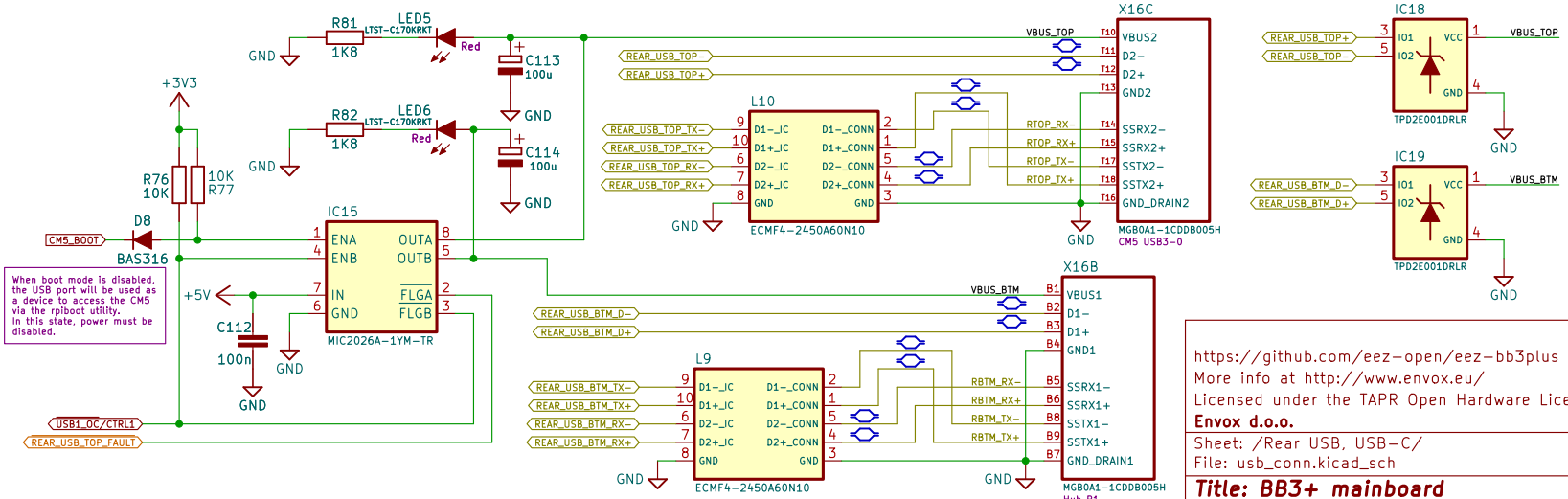
USB-C 3.0 front port (source max 1.5A)



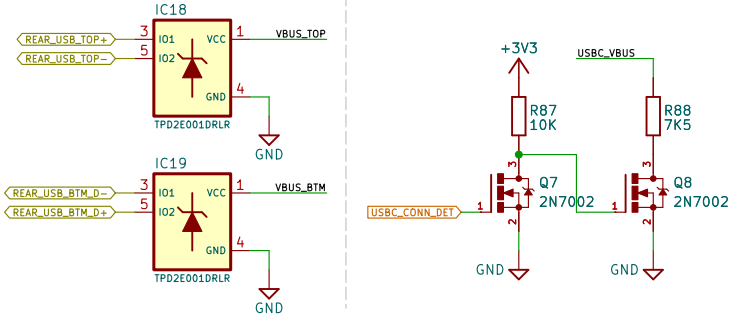
USB-C 3.0 Protection and Mux



USB2.0 Type-A (Rear panel)



VBus Discharge Circuit



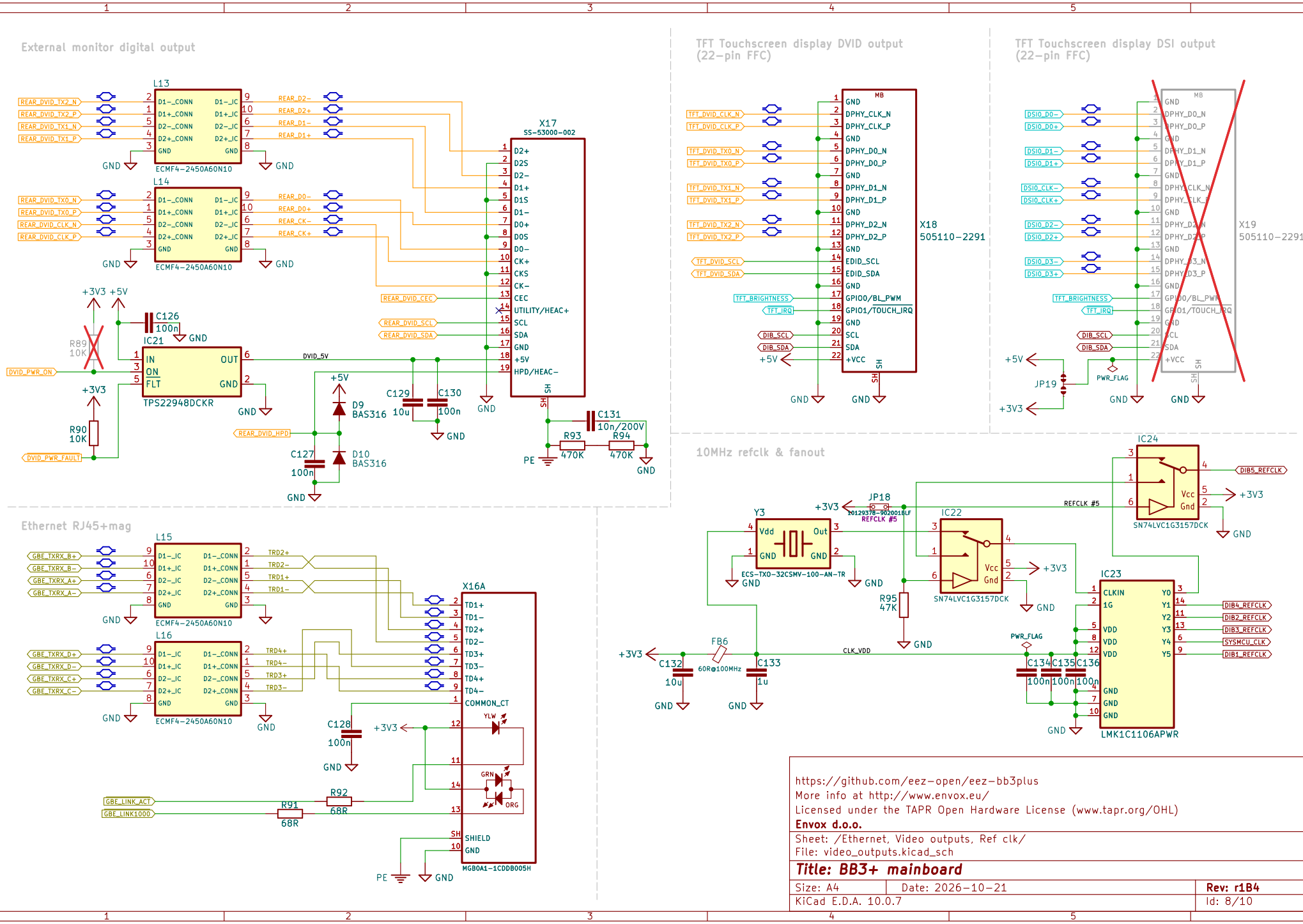
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

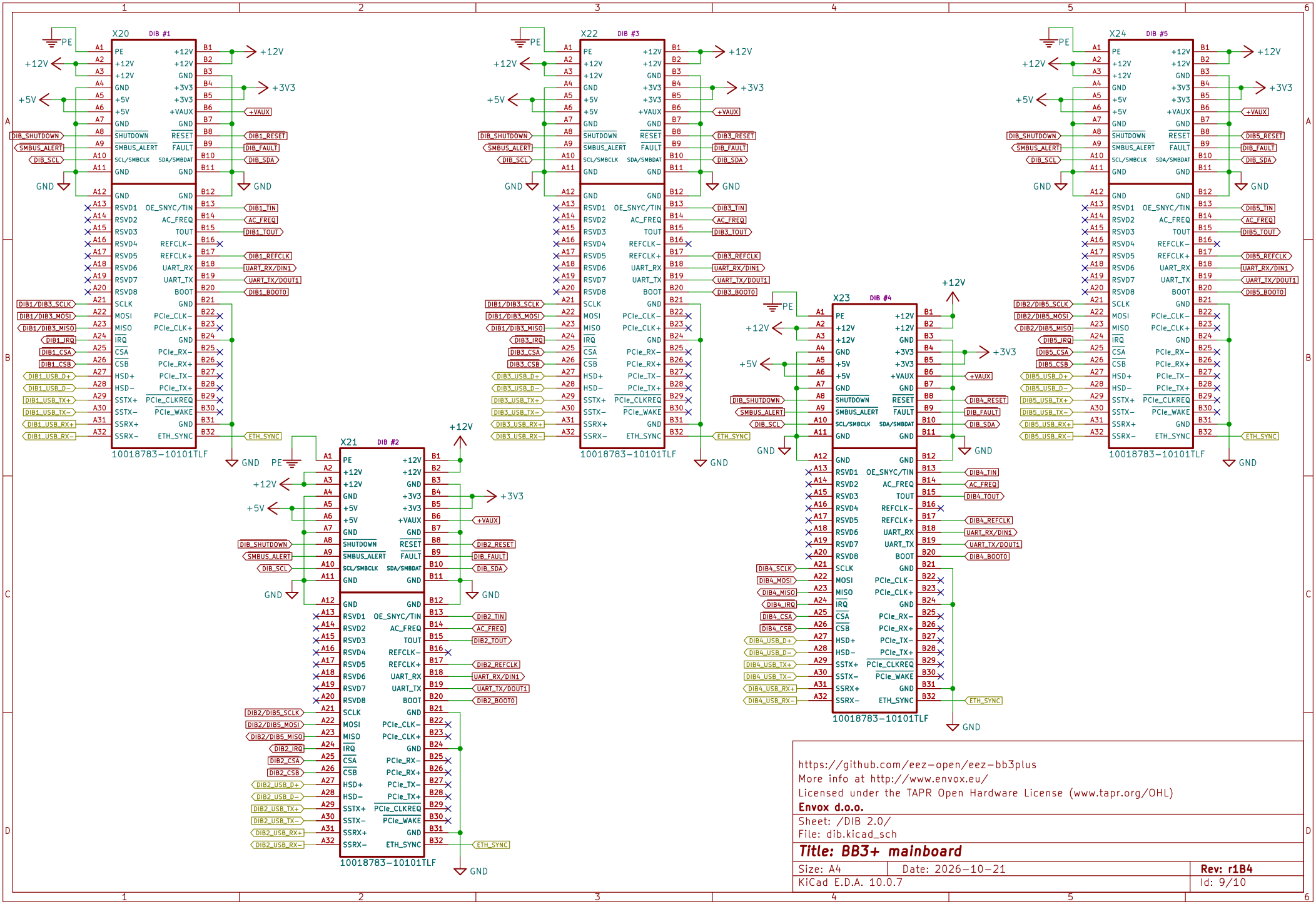
Envox d.o.o.

Sheet: /Rear USB, USB-C/
File: usb_conn.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-10-21	**Rev: r1B4**
KiCad E.D.A. 10.0.7		Id: 7/10

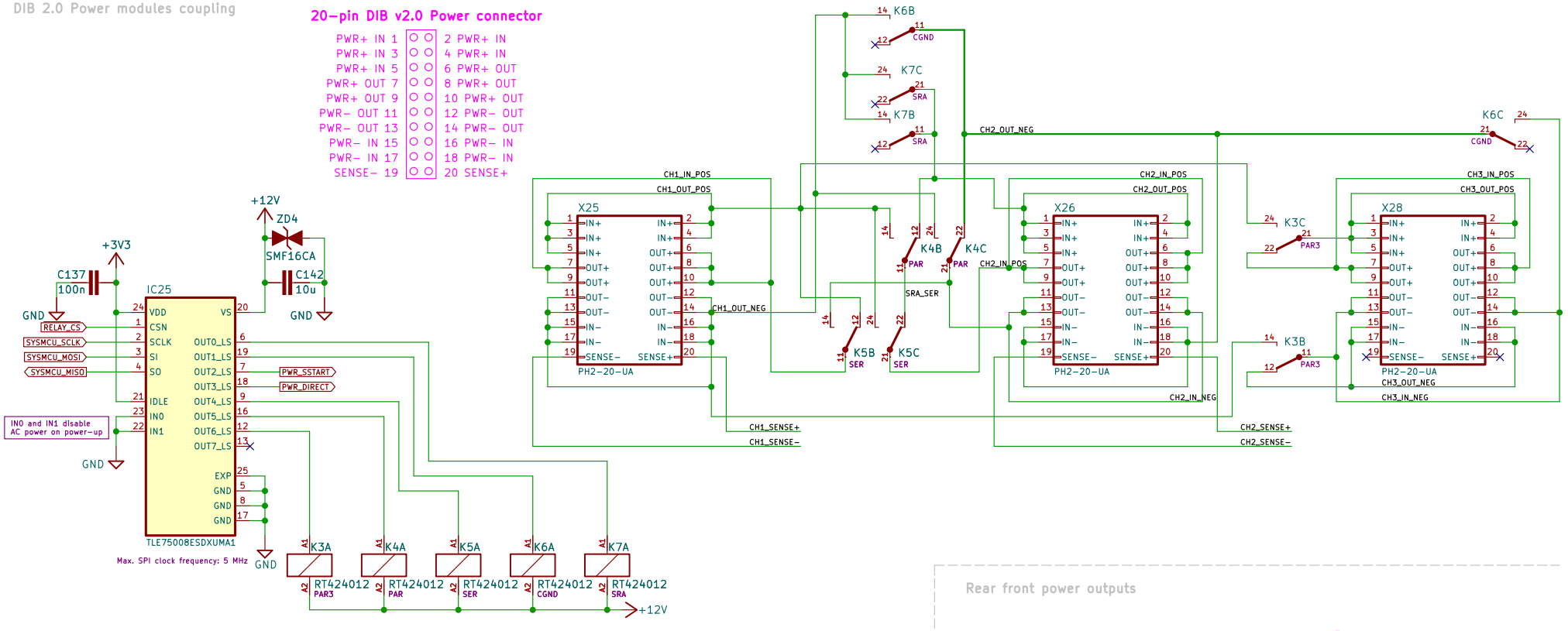

External monitor digital output
L13
ECMF4-2450A60N10
L14
ECMF4-2450A60N10
X17
SS-53000-002
IC21
TPS22948DCKR
C126
100n
R89
10K
R90
10K
C129
10u
C130
100n
C127
100n
D9
BAS316
D10
BAS316
C131
10n/200V
R93
470K
R94
470K
TFT Touchscreen display DVID output
(22-pin FFC)
X18
505110-2291
TFT Touchscreen display DSI output
(22-pin FFC)
X19
505110-2291
JP19
PWR_FLAG
Ethernet RJ45+mag
L15
ECMF4-2450A60N10
L16
ECMF4-2450A60N10
X16A
MGB0A1-1CDDB005H
C128
100n
R91
68R
R92
68R
10MHz refclk & fanout
IC24
SN74LVC1G3157DCK
JP18
2129378-9020018LF
Y3
ECS-TXO-32CSMV-100-AN-TR
IC22
SN74LVC1G3157DCK
R95
47K
FB6
60R@100MHz
C132
10u
C133
1u
IC23
LMK1C1106APWR
C134
100n
C135
100n
C136
100n
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.
Sheet: /Ethernet, Video outputs, Ref clk/
File: video_outputs.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-21
Rev: r1B4
KiCad E.D.A. 10.0.7
Id: 8/10


X20
DIB #1
X21
DIB #2
X22
DIB #3
X23
DIB #4
X24
DIB #5
10018783-10101TLF
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.
Sheet: /DIB 2.0/
File: dib.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-21
Rev: r1B4
KiCad E.D.A. 10.0.7
Id: 9/10

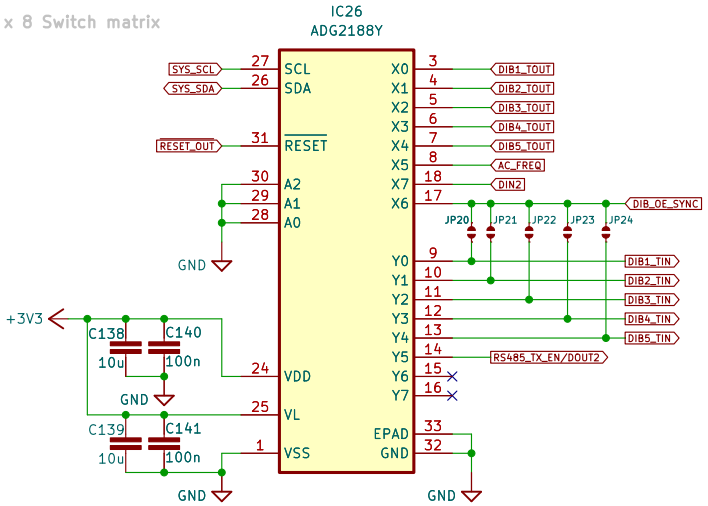
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

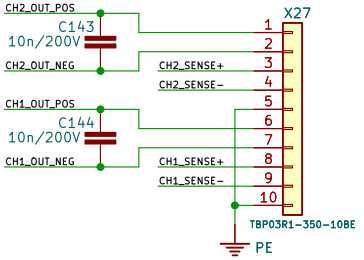
PWR+ IN 1	○ ○	2	PWR+ IN
PWR+ IN 3	○ ○	4	PWR+ IN
PWR+ IN 5	○ ○	6	PWR+ OUT
PWR+ OUT 7	○ ○	8	PWR+ OUT
PWR+ OUT 9	○ ○	10	PWR+ OUT
PWR- OUT 11	○ ○	12	PWR- OUT
PWR- OUT 13	○ ○	14	PWR- OUT
PWR- IN 15	○ ○	16	PWR- IN
PWR- IN 17	○ ○	18	PWR- IN
SENSE- 19	○ ○	20	SENSE+



Trigger 8 x 8 Switch matrix



Rear front power outputs



Rear panel power connector

(Slots #1 and #2 only)

CH2_OUT+ 1	○ ○	2	CH2_OUT-
CH2_SENSE+ 3	○ ○	4	CH2_SENSE-
PE 5	○ ○	6	CH1_OUT+
CH1_OUT- 7	○ ○	8	CH1_SENSE+
CH1_SENSE- 9	○ ○	10	PE

https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.

Sheet: /Trigger switch matrix, Power modules coupling/
File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-10-21	**Rev: r1B4**
KiCad E.D.A. 10.0.7		Id: 10/10